LAB 3 Tournament Management System

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Architecture Selection: We chose Component-Based Architecture for the Tournament Management System.

1. Architectural Choice

The system uses a Component-Based Architecture in which the main functionality is divided into separate subsystems and components: Team Management, Match Management, and Tournament Management. These components communicate through defined ports and interfaces.

2. Two Reasons for the Choice

- **Modular functionality:** Team registration and validation, match result handling, and tournament creation/bracket generation are separated into independent components. This makes the Tournament Management System easier to develop and maintain.
- **Clear communication between modules:** The subsystems exchange information such as Registered Teams, Match Information, Updated Results, and Tournament Details through ports/interfaces. This reduces direct dependency between internal components and supports easier modification.

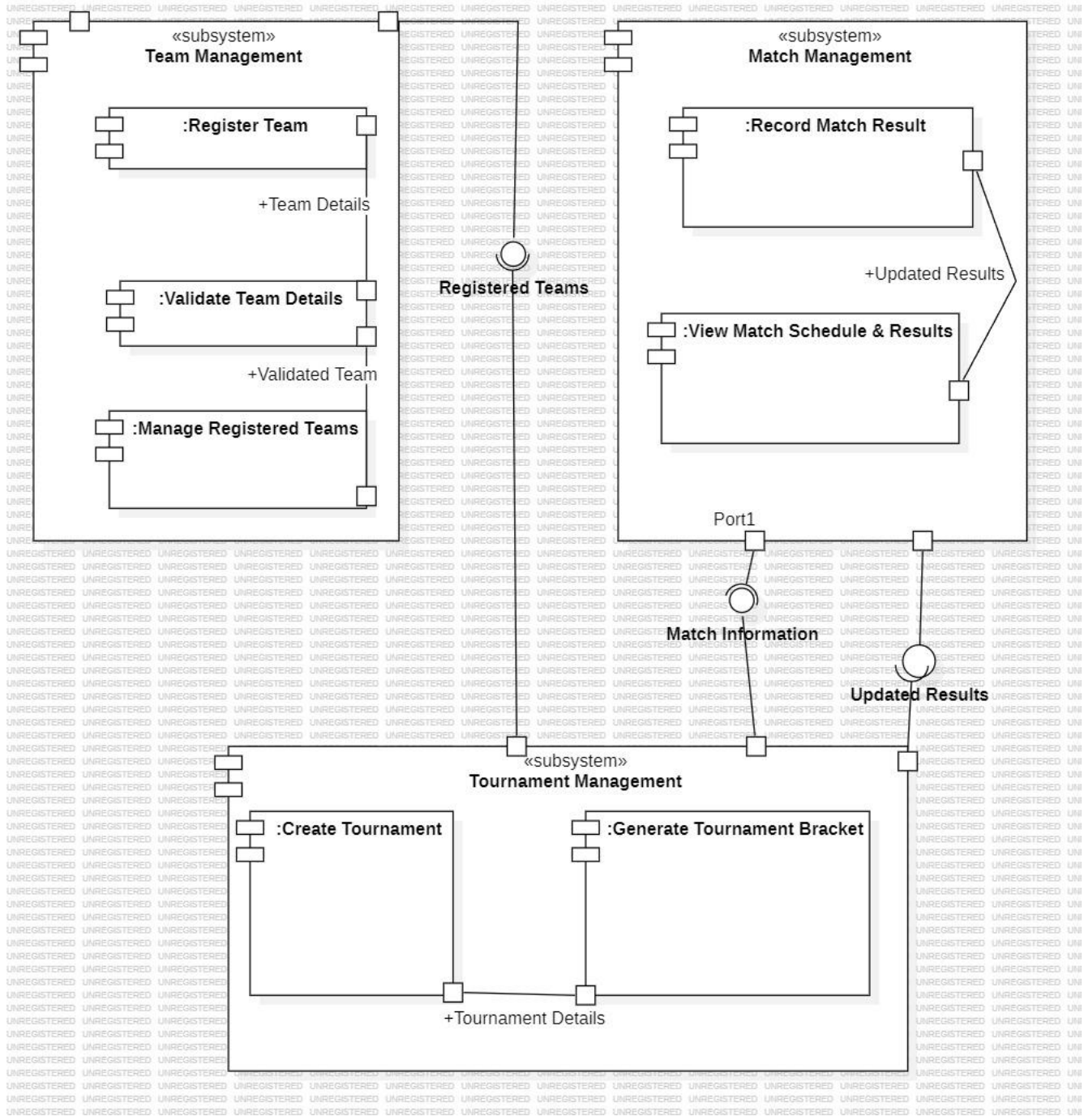
3. Security Advantage

The architecture exposes functionality through defined interfaces rather than giving every component direct access to another component's internal operations. For example, only the required team or match information is exchanged through the corresponding interfaces. This helps enforce controlled access between subsystems and reduces unnecessary exposure.

4. Performance Benefit

Independent components can handle different operations without requiring the entire system to be processed as one unit. For example, team management and match-related processing can operate independently when their required data is available. This reduces unnecessary processing and can improve overall system responsiveness.

5. Diagram


«subsystem»
Team Management
:Register Team
+Team Details
:Validate Team Details
+Validated Team
:Manage Registered Teams
Registered Teams
«subsystem»
Match Management
:Record Match Result
+Updated Results
:View Match Schedule & Results
Port1
Match Information
Updated Results
«subsystem»
Tournament Management
:Create Tournament
:Generate Tournament Bracket
+Tournament Details